Z - test

1) A simple random sampling survey in respect of monthly earnings of semi-skilled workers in two cities gives the following statistical information.

City	Mean monthly earnings (Rs)	Std. deviation of sample data of monthly earnings	Size of sample
A	695	40	200
B	710	60	175

Test the hypothesis at 5% level that there is no difference btw monthly earnings of workers in 2 cities.

Step 1 :- Form hypothesis :-

H_0 : There is no difference btw monthly earnings of workers in 2 cities.

H_1 : There is a difference btw monthly earnings of workers in 2 cities.

Step 2 :- Select test

As the sample size is large and variance is known, use z test

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_A^2}{n_A} + \frac{\sigma_A^2}{n_B}}}$$

Step 3 :- Z - Statistics

$$Z = \frac{695 - 710}{\sqrt{\frac{(40)^2}{200} + \frac{(60)^2}{175}}} = -2.809$$

$\therefore$ Calculated value of $Z = -2.809$

Step 4 :- Find the tabulated value of Z-statistic

Significance level is 5% = 0.05
H_1 is 2-tailed. i.e., (H_1 is $\mu_1 \neq \mu_2$) which means both greater than or less than is rejected.

2) A company manufactures resistors with a claimed mean resistance of 100Ω. The population standard deviation is $\sigma = 2.5\Omega$. A random sample of $n = 60$ resistors is taken, with sample mean $\bar{x} = 99.4\Omega$. Find (a) At $\alpha = 0.01$, test if the true mean differs from 100 Ω
(b) Construct the 99% confidence interval for μ.

Given $\mu_0 = 100$ → population mean

S.D $\sigma = 2.5$

sample mean $= \bar{x} = 99.4$

Sample size $n = 60$

significance level $= 0.01$

(a) Hypothesis $H_0 : \mu = 100$

$H_1 : \mu \neq 100$

Two tailed

$$Z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = 1.86$$

Critical value:- $Z_{0.005} = \pm 2.58$

$|z| = 1.86 < 2.58$ ⇒ fail to reject H_0

(b) Confidence interval (99%) $\left\{ \begin{array}{l} C = 0.99 \Rightarrow \\ \alpha : 1 - 0.99 = 0.01 \end{array} \right\}$

$$C.I = \bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 99.4 \pm 2.58 \times 0.3227$$

$$= (98.57, 100.23)$$

Conclusion:- No significance evidence that mean differs from 100

3) A coffee shop claims that the average amount of coffee in its cups is 250 ml. A random sample of 36 cups is measured and the sample mean is found to be 247 ml. The population s.d is known: $\sigma = 5$ ml. Find.

(a) At $\alpha = 0.05$, test weather the average amount differs from 250 ml.

(b) Construct a 95% confidence interval for the mean coffee amount.

$\mu_0 = 250$ ml $\qquad n = 36$

$\bar{x} = 247$ ml $\qquad \alpha = 0.05$

$\sigma = 5$ ml

(a) Hypothesis:-

$H_0 = \mu = 250$

$H_1 : \mu \neq 250$ (2 tailed).

$$Z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = -3.60$$

$Z_{\alpha/2} = \pm 1.96$ (From z table)

$|Z| = 3.60 > 1.96 \Rightarrow$ Reject H_0

(b) $CI = \bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

$= 247 \pm 1.96 \left(\frac{5}{\sqrt{36}}\right)$

$c = 0.95$

$\Rightarrow \alpha = 1 - 0.95$

$= 0.05$

$\therefore \boxed{Z_\alpha = 1.96}$

$= (245.37,\ 248.63)$

This interval doesnot include 250 ml. The coffee shop's claim of 250 ml is not supported by this sample.

The average amount of coffee in the cups is significantly less than 250 ml.

4) Two battery brands, Brand A and Brand B, are claimed to have equal mean lifetimes. Population standard deviations are known

$\sigma_A = 3.5$ hrs, $\sigma_B = 4.0$ hrs.

Samples : Brand A : $n_A = 40$, $\bar{X}_A = 41.8$ hrs

Brand B : $n_B = 50$, $\bar{X}_B = 39.6$ hrs

Find

(a) At $\alpha = 0.01$, test if the mean difference differ significantly.

(b) Construct a 95% confidence interval for $(\mu_A - \mu_B)$

$\bar{X}_A = 41.8$ $\quad$ $\bar{X}_B = 39.6$

$\sigma_A = 3.5$ $\quad$ $\sigma_B = 4.0$

$n_A = 40$ $\quad$ $n_B = 50$

$\alpha = 0.01$

(a) Hypothesis :- $H_0 = \mu_A = \mu_B$

$H_1 = \mu_A \neq \mu_B$ (Two - tailed)

$$Z = \frac{\bar{X}_A - \bar{X}_B}{\sqrt{\frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}}} = \frac{41.8 - 39.6}{\sqrt{\frac{3.5^2}{40} + \frac{4^2}{50}}} = \frac{2.2}{0.79136}$$

$$Z = 2.78$$

$$Z_{0.005} = 2.58$$

$$|Z| = 2.78 > 2.58 \Rightarrow \text{Reject } H_0$$

(b) C.I at 95% $\Rightarrow C = 0.95 \Rightarrow \alpha = 1 - 0.95 = 0.05$

$$\therefore Z_{\alpha/2} = 1.96$$

$$C.I = \bar{X}_A - \bar{X}_B \pm Z_{\alpha/2} \sqrt{\frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}}$$

$$= 2.2 \pm (1.96) \sqrt{0.62625}$$

$$= (0.649, 3.751)$$

$\Rightarrow$ Brand A lasts significantly longer than brand B

5) A factory claims that its light bulbs last at least 1200 hrs on average. A random sample of 40 bulbs is tested, and the sample mean lifetime is 1185 hrs.

The population S.D is 30 hrs.
Find at $\alpha = 0.05$, test whether the mean lifetime is less than 1200 hrs.

$n = 40$ $\quad\quad \bar{x} = 1185$

$\sigma = 30$ hrs $\quad\quad \mu_0 = 1200$

Hypothesis :-

$H_0 : \mu \geq 1200$

$H_1 : \mu < 1200$ (one tailed)

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{1185 - 1200}{30/\sqrt{40}} = \frac{-15}{4.743} = -3.16$$

$Z_\alpha = Z_{0.05} = -1.645$ (left tailed)

$|Z| = 3.16 > |Z_\alpha| = 1.645 \Rightarrow$ Reject H_0

$\therefore$ The factory's claim of at least 1200 hours is not supported by this sample.

t-test

1) A process scheduler claims mean CPU utilization = 70%. Sample of 10 runs are 68, 72, 69, 70, 65, 71, 68, 74, 69, 67

At $\alpha = 0.05$, verify claim.

H_0 : mean CPU utilization = 70%

H_1 : mean CPU utilization ≠ 70%

$$\bar{x} = \frac{68+72+69+70+65+71+68+74+69+67}{10}$$

$= 69.3$

x	$x-\bar{x}$	$(x-\bar{x})^2$
68	-1.3	1.69
72	2.7	7.29
69	-0.3	0.09
70	0.7	0.49
65	-4.3	18.49
71	1.7	2.89
68	-1.3	1.69
74	4.7	22.09
69	-0.3	0.09
67	-2.3	5 29

$$\boxed{\Sigma (x-\bar{x})^2 = 60.1}$$

$$S.D = S = \sqrt{\frac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{n-1}} = \sqrt{\frac{60.1}{9}} = 2.58$$

$$t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} = \frac{69.3 - 70}{2.58/\sqrt{10}}$$

$t = -0.858 = t_{cal}$

degree of freedom $= n-1 = 10-1 = 9$

$t_{0.05,9} = 2.262 = t_{tab}$

$|t_{cal}| = 0.858 < 2.262 = t_{tab}$

$\Rightarrow$ fail to reject H_0

$\Rightarrow$ Mean CPU utilization = 70%

2. Compare two ML model's accuracies (%) on 10-fold CV:

Model A = [91, 89, 90, 92, 88, 93, 90, 91, 89, 90]

Model B = [88, 85, 87, 86, 88, 89, 86, 87, 85, 86]

Test whether there is a significant difference in mean accuracies at $\alpha = 0.05$ (assume equal variance)

H_0 = There is no significant difference in mean accuracies

H_1 = There is significant difference in mean accuracies

$$\bar{X}_A = \frac{91+89+90+92+88+93+90+91+89+90}{10} = 90.3$$

$$\bar{X}_B = \frac{88+85+87+86+88+89+86+87+85+86}{10} = 86.7$$

Sample A

X_A	$X_A - \bar{X}_A$	$(X_A - \bar{X}_A)^2$
91	0.7	0.49
89	-1.3	1.69
90	-0.3	0.09
92	1.7	2.89
88	-2.3	5.29
93	2.7	7.29
90	-0.3	0.09
91	0.7	0.49
89	-1.3	1.69
90	-0.3	0.09

$$S_A = \sqrt{\frac{\sum (X_A - \bar{X}_A)^2}{n-1}}$$

$$= \sqrt{\frac{20.10}{9}} = 1.49$$

Sample B

X_B	$X_B - \bar{X}_B$	$(X_B - \bar{X}_B)^2$
88	1.3	1.69
85	-1.7	2.89
87	0.3	0.09
86	-0.7	0.49
88	1.3	1.69
89	2.3	5.29
86	-0.7	0.49
87	0.3	0.09
85	-1.7	2.89
86	-0.7	0.49

$$S_B = \sqrt{\frac{(X_B - \bar{X}_B)^2}{n-1}}$$

$$= \sqrt{\frac{16.10}{9}} = 1.34$$

It is given equal variance.

$$\therefore \quad S_p = \sqrt{\frac{(n_A-1)S_A^2+(n_B-1)S_B^2}{n_A+n_B-2}} = \sqrt{\frac{9\times1.49^2+9\times1.34^2}{18}}$$

$$S_p = 1.417$$

$$t = \frac{\bar{X}_A-\bar{X}_B}{S_p\sqrt{\frac{1}{n_A}+\frac{1}{n_B}}} = \frac{90.3-86.7}{1.417\sqrt{\frac{1}{10}+\frac{1}{10}}} = 5.6809$$

$$df = n_1+n_2-2 = 18$$

$$t_{0.05,\,18} = 2.101$$

$$t_{cal} = 5.6809 > 2.101 = t_{tab}$$

$\Rightarrow$ Reject H_0